# Classification Is All You Need


## Abstract

Language models are widely used for classification, but generating an answer does not directly provide a probability distribution over a fixed label set. We evaluate Jev, TypeSafe's first System One model, zero-shot on four public datasets against zero-shot Qwen3.6-27B self-hosted on an AWS SageMaker `ml.g6.12xlarge` instance (four NVIDIA L4 GPUs), BERT fine-tuned on an `ml.g5.xlarge` instance (one NVIDIA A10G GPU), and XGBoost trained on an `ml.m5.xlarge` CPU instance. The comparison measures accuracy, inference cost, latency, and training-label requirements. On balanced or nearly balanced text tasks evaluated with equal error costs, Jev reaches 96.0% accuracy on SST-2 and 87.9% on AG News, matching Qwen (95.6% and 87.7%) at substantially lower inference cost and latency. Across the four datasets, Jev costs \$0.015–\$0.030 per 1,000 predictions and takes 0.14–0.15 seconds per request, compared with \$0.352–\$1.342 in AWS compute and 0.92–1.10 seconds for Qwen. On AG News, BERT and XGBoost surpass both zero-shot models with 5,752 and 28,752 training examples, respectively—the smallest tested budgets, alongside validation labels for model selection. Neither surpasses Jev on SST-2 with all 65,349 training examples. Once trained, supervised models offer orders-of-magnitude cheaper inference, with microsecond-to-millisecond batch time per item. On the imbalanced German Credit and Breast Cancer Wisconsin datasets, XGBoost achieves the highest default macro-F1 scores, at 0.657 and 0.945, respectively. Jev produces the fewest false negatives but substantially more false positives. This tradeoff warrants evaluation in applications where missed positive cases are especially costly, but model selection requires explicit error costs and comparison at suitable decision thresholds. All Jev results are zero-shot, using one fixed input template per dataset without prompt optimization. Finally, we evaluate KernelSHAP and LIME, showing that these standard perturbation-based explanation methods apply directly to Jev's class probabilities and identify words or features that influence its predictions.


## 1 Introduction

Much of practical machine learning is classification: approve or reject a loan, route a ticket, flag a transaction. The discipline that grew around it assumes a model that outputs a probability for each class. Calibration diagnostics, cost-sensitive decision rules, label-efficient evaluation and model-agnostic explanations all consume that distribution.

Large language models are now routinely used as zero-shot classifiers, including on tabular data serialized to text [8]. But a language model does not output a class distribution. It defines a measure over token sequences, and a label is recovered only after parsing its output. Mass leaks to strings that name no class, and the class distribution can at best be estimated by sampling or by asking the model to verbalize numbers, which are coarse and often poorly calibrated [10, 21, 26]; its self-reported explanations need not reflect what drove the prediction [22].

Some models take a different route: given an input and a predefined set of options, they return a probability for *every* option [23]. Our thesis is that this is not a cosmetic difference. Moving the output from a measure on strings to a measure on the finite class space $(C, 2^C)$ is what the classical toolbox requires, including the part of it whose job is to say when a model should not be trusted.

We contribute (i) a short formal account of the measure a generative model induces on classes and of the invalid mass it leaks (Section 2); (ii) an evaluation of one such model, Jev, zero-shot on

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four public datasets, next to an open-weight language model scored so that it also returns a class distribution and to two supervised models, on identical examples and with every label budget stated (Sections 4–5); (iii) a comparison of macro-F1, cost and latency against the number of training labels, and an ensemble that escalates low-confidence predictions (Sections 6–7); and (iv) threshold selection, importance-sampled evaluation and perturbation-based explanations driven only by the class probabilities (Sections 8–10).

The tools do not flatter the model. They show that its probabilities are accurate and close to calibrated on text and biased on the two tabular tasks, in one case in a way a threshold repairs and in the other not; that a textbook importance-sampling proposal does not pay off on these probabilities, and why; that a self-hosted open model matches it on text at many times the price per prediction; and that escalating its uncertain predictions helps exactly where its confidence is informative. The paper is about which tooling a class distribution makes available and what that tooling reveals. It is not a leaderboard: two models are zero-shot and two are supervised.

## 2 From a measure on strings to a measure on classes

Let $\Sigma$ be a finite vocabulary and $\Sigma^*$ the countable set of finite token sequences with its discrete $\sigma$-algebra $2^{\Sigma^*}$. Given an input $x$, an autoregressive language model that terminates almost surely defines a probability measure $P_x$ on $(\Sigma^*, 2^{\Sigma^*})$, $P_x(\{s\}) = \prod_{t=1}^{|s|} p_\theta(s_t \mid s_{<t}, x)\, p_\theta(\texttt{eos} \mid s, x)$. Using it as a classifier over a finite label set $C$ requires a *parsing map* $\pi : \Sigma^* \to C \cup \{\bot\}$ that sends each string to the label it names, or to $\bot$ when it names none.

**Definition 1** (Induced class measure and invalid mass)**.** *The measure the model induces on labels is the pushforward $Q_x = P_x \circ \pi^{-1}$ on $(C \cup \{\bot\},\, 2^{C \cup \{\bot\}})$, with $Q_x(\{c\}) = \sum_{s:\, \pi(s)=c} P_x(\{s\})$. The* invalid mass *$\delta(x) = Q_x(\{\bot\})$ is the probability that the answer is not a class at all; conditioning on validity gives $q_x(c) = Q_x(\{c\})/(1 - \delta(x))$ when $\delta(x) < 1$.*

Two obstacles separate $P_x$ from a usable class distribution. Each preimage $\pi^{-1}(c)$ is an infinite set of strings, so $Q_x(\{c\})$ cannot be evaluated exactly: one reads the probability of a label's first token, samples $k$ completions (variance $q_x(c)(1 - q_x(c))/k$, resolution $1/k$), or asks for verbalized numbers, which are strings with their own invalid mass. And $\delta(x)$ is an input-dependent loss of mass that every downstream procedure must handle.

A classification model instead outputs a probability measure $\mu_x$ on $(C, 2^C)$ directly: a vector $(\mu_x(\{c\}))_{c \in C}$ summing to one, with $\delta(x) \equiv 0$ by construction. The tools we evaluate all take $\mu_x$ as input and need more than an argmax label. Proper scoring rules and calibration (Brier score, log-loss, ECE) evaluate $\mu_x(\{c\})$ for every $c$ [1, 6, 15]. Ranking metrics and threshold selection use the positive-class probability as a score, and a cost-sensitive decision picks an operating point analytically [5] or from validation labels. Label-efficient evaluation by importance sampling uses the predicted uncertainty $1 - \max_c \mu_x(\{c\})$ to choose which items to label [11, 18]. Perturbation-based attribution needs a real-valued function $f(x) = \mu_x(\{c^\star\})$ on many perturbed inputs [13, 17]. With a generative model each of these consumes an estimate $\hat{q}_x$ that carries sampling noise and the invalid mass and multiplies the number of model calls.

## 3 Related evaluations of Jev

Jev has been evaluated publicly on three of our four datasets. Deußer et al. [3] benchmark the same model version zero-shot on 37 datasets and report that its choice probabilities are well calibrated on text, but that binary probabilities "rank well but are poorly placed relative to a fixed 0.5 threshold"; a public repository reports 500-item samples of the text datasets [14], and another evaluates Breast Cancer Wisconsin with its own serialization [20]. We found no public evaluation on German Credit. Our text results agree with these (Appendix A, Table 9). None of these studies trains supervised models on the same examples, reports the exact requests sent, or applies cost-sensitive thresholds, importance-sampled evaluation, escalation or perturbation-based explanations.

Table 1: Datasets. Evaluation examples are the official held-out split (text) or the test part of each of 10 split repeats (tabular); training and validation sizes are per repeat.

| Dataset | Type | Classes | Evaluation | Training | Validation |
|---|---|---|---|---|---|
| German Credit | tabular | 2 | $10 \times 200$ | 600 | 200 |
| Breast Cancer Wisconsin | tabular | 2 | $10 \times 114$ | 341 | 114 |
| SST-2 | text | 2 | 872 | 65,349 | 2,000 |
| AG News | text | 4 | 7,600 | 115,000 | 5,000 |

# 4 Setup

**Datasets and protocol.** We use four public datasets (Table 1): German Credit [9] and Breast Cancer Wisconsin Diagnostic [25], two small binary tabular tasks, SST-2 sentiment [19, 24] and four-class AG News [27]. The text datasets are evaluated on their complete official held-out split, with training and validation data taken from the official training split only. The tabular datasets are too small for one split to be reliable, so they use 10 seeded, stratified 60/20/20 train/validation/test repeats. All identifiers were frozen, with a content hash, before any template check, training or threshold selection.

The two kinds of dataset pose different problems. The text test sets are balanced (428 negative and 444 positive SST-2 sentences; 1,900 AG News articles in each class) and we treat their errors as equally costly, so accuracy is a fair summary. The tabular datasets are small and imbalanced (30% of applicants are bad, 37% of masses malignant), and their errors are not equally costly: missing a bad applicant or a malignant mass is worse than a false alarm. For them we therefore report the counts of each kind of error and a decision cost in which a miss costs five times a false alarm (canonical for German Credit, assumed for Breast Cancer Wisconsin). Held-out datasets get 95% bootstrap intervals; tabular datasets get the mean and standard deviation over the repeats (Appendix B).

**Jev, zero-shot.** Jev (version `jev-1.13.0`) [23] is queried with one question per example: a one-sentence instruction and one description per class, in a fixed order. Tabular rows are serialized as one `name: value` line per feature; text is sent as is. Each dataset has one template, frozen without tuning. Appendix I shows complete requests and responses.

**Qwen3.6-27B, zero-shot.** The open-weight `Qwen/Qwen3.6-27B` [16] is self-hosted and receives exactly Jev's content: its instruction, options and descriptions in the same order, then the serialized input. To obtain a class distribution rather than a generated string, every option is scored as the answer: the log-probabilities of the tokens of the option name and of the end-of-turn token are summed, and a softmax over the options gives $\mu_x$ (Appendix J). Thinking is disabled, except on 50 items per dataset reported separately.

**XGBoost and BERT, supervised.** XGBoost [2] uses one-hot tabular features or TF-IDF text features; a `bert-base` encoder [4] with a classification head is fine-tuned on the text Jev receives, so a tabular row is its serialized lines. Both select their configuration on validation log-loss only, are trained without evaluation labels, and are refitted on nested fractions of the training part for learning curves. No model is recalibrated.

**Label budgets and cost.** Jev and Qwen use no training labels and use validation labels only where a decision or escalation threshold is selected; XGBoost and BERT use every training and validation label (Appendix B, Table 10). All Jev requests cost USD 1.64 and all cloud training and inference jobs USD 25.67; a fifth dataset was run and then removed from the study, and its share of both is stated in Appendix B.

# 5 Classical metrics and calibration

Because every model returns a full distribution, one metric toolbox scores all four (Table 2); no Jev prediction was invalid. Appendix C has the full metrics with PR-AUC (Table 12), Brier score and log-loss (Table 13) and reliability diagrams (Figure 4); Appendix L has per-class metrics.

Table 2: Zero-shot (Jev, Qwen3.6-27B) and supervised (XGBoost, BERT) models on identical evaluation examples. Accuracy carries a 95% bootstrap interval (text) or the standard deviation over 10 split repeats (tabular). ROC-AUC is for the positive class (binary) or one-vs-rest macro (AG News). Label budgets differ (Table 10).

| Dataset | Model | Accuracy | Macro-F1 | ROC-AUC | ECE |
|---|---|---|---|---|---|
| German Credit | Jev | 0.467 $\pm$0.019 | 0.467 | 0.537 | 0.364 |
| | Qwen3.6-27B | 0.544 $\pm$0.023 | 0.457 | 0.496 | 0.136 |
| | XGBoost | 0.741 $\pm$0.031 | 0.657 | 0.768 | 0.069 |
| | BERT | 0.705 $\pm$0.008 | 0.473 | 0.683 | 0.060 |
| Breast Cancer Wisconsin | Jev | 0.411 $\pm$0.014 | 0.343 | 0.917 | 0.374 |
| | Qwen3.6-27B | 0.825 $\pm$0.020 | 0.783 | 0.969 | 0.072 |
| | XGBoost | 0.949 $\pm$0.021 | 0.945 | 0.985 | 0.036 |
| | BERT | 0.778 $\pm$0.092 | 0.716 | 0.853 | 0.115 |
| SST-2 | Jev | 0.960 [0.948, 0.971] | 0.960 | 0.987 | 0.020 |
| | Qwen3.6-27B | 0.956 [0.943, 0.968] | 0.956 | 0.989 | 0.023 |
| | XGBoost | 0.791 [0.763, 0.819] | 0.791 | 0.873 | 0.026 |
| | BERT | 0.916 [0.896, 0.935] | 0.916 | 0.976 | 0.054 |
| AG News | Jev | 0.879 [0.871, 0.886] | 0.878 | 0.961 | 0.082 |
| | Qwen3.6-27B | 0.877 [0.870, 0.885] | 0.877 | 0.962 | 0.099 |
| | XGBoost | 0.914 [0.908, 0.921] | 0.914 | 0.985 | 0.032 |
| | BERT | 0.942 [0.936, 0.947] | 0.942 | 0.992 | 0.018 |

Table 3: Error counts on the two tabular datasets, summed over the test parts of the 10 split repeats. Positive is *bad* (German Credit) or *malignant* (Breast Cancer Wisconsin); cost is $5\,\text{FN} + \text{FP}$. Left: the default decision, positive when $P(\text{positive}) \geq 0.5$. Right: the threshold selected on validation labels to minimize this cost (Section 8). The last row of each block calls every case positive.

| | | Default decision | | | | | Validation-selected threshold | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Dataset | Model | TP | FN | FP | TN | Cost | TP | FN | FP | TN | Cost |
| German Credit | Jev | 474 | 126 | 945 | 455 | 1,575 | 576 | 24 | 1,275 | 125 | 1,395 |
| | Qwen3.6-27B | 151 | 449 | 467 | 933 | 2,712 | 576 | 24 | 1,306 | 94 | 1,426 |
| | XGBoost | 248 | 352 | 166 | 1,234 | 1,926 | 523 | 77 | 747 | 653 | 1,132 |
| | BERT | 56 | 544 | 48 | 1,352 | 2,768 | 551 | 49 | 1,034 | 366 | 1,279 |
| | *flag every case* | 600 | 0 | 1,400 | 0 | 1,400 | 600 | 0 | 1,400 | 0 | 1,400 |
| Breast Cancer Wisconsin | Jev | 417 | 3 | 675 | 45 | 690 | 383 | 37 | 265 | 455 | 450 |
| | Qwen3.6-27B | 223 | 197 | 2 | 718 | 987 | 389 | 31 | 108 | 612 | 263 |
| | XGBoost | 386 | 34 | 24 | 696 | 194 | 396 | 24 | 48 | 672 | 168 |
| | BERT | 249 | 171 | 82 | 638 | 937 | 355 | 65 | 267 | 453 | 592 |
| | *flag every case* | 420 | 0 | 720 | 0 | 720 | 420 | 0 | 720 | 0 | 720 |

**Text.** Zero-shot Jev is accurate and its probabilities are usable as they come: accuracy 0.960 on SST-2 and 0.879 on AG News, with ECE 0.020 and 0.082, close to prior public evaluations of the same model version. Qwen is indistinguishable from it in macro-F1 (0.956 and 0.877 against 0.960 and 0.878). Jev reports probabilities rounded to two decimals, so a true class can receive a reported probability of exactly zero; this dominates its log-loss on AG News and matters again in Sections 7 and 9.

**Tabular.** Here Jev's default decision fails. It labels 0.70 of German Credit applicants *bad* where 0.30 are, and 0.95 of breast masses *malignant* where 0.37 are, so its accuracy (0.467 and 0.411) is below the majority-class rate, with ECE 0.364 and 0.374. The two failures differ, and only a class distribution tells them apart. On Breast Cancer Wisconsin the ranking is good (ROC-AUC 0.917): the probabilities carry the signal and are misplaced relative to 0.5, which a threshold corrects (Section 8). On German Credit the ranking itself is weak (ROC-AUC 0.537). Qwen does not share the bias on Breast Cancer Wisconsin: it calls 0.20 of the masses malignant and reaches macro-F1 0.783 against Jev's 0.343, and it is better calibrated on both tables (ECE 0.136 and 0.072). XGBoost is the strongest

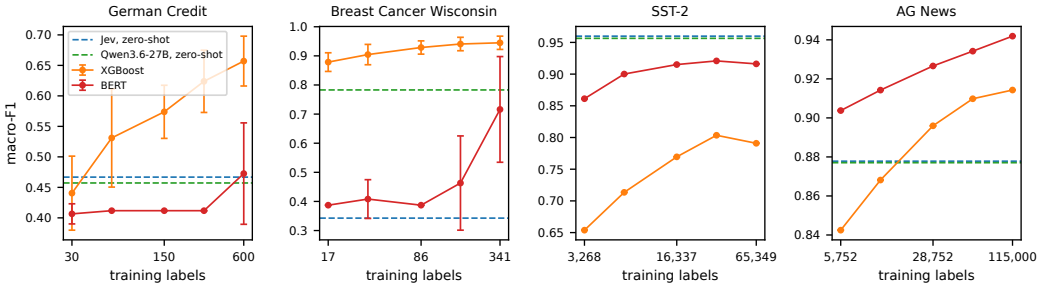



Figure 1: Macro-F1 on the evaluation examples against the number of training labels. Curves: supervised models. Dashed lines: zero-shot models, which use no training labels.

Table 4: Labels, cost in USD and latency. Training cost is the whole recorded job, including model selection and the learning curve. Cost per 1,000 predictions: Jev, mean recorded request cost; Qwen, the billed scoring job allocated by scoring time, with inference time only in parentheses; XGBoost and BERT, measured batch inference time at list price. Latency: Jev, one API request; Qwen, one item sent alone, with batch throughput in parentheses; XGBoost and BERT, batch inference time per item.

| Dataset | Model | Labels | Macro-F1 | Training | Per 1,000 | Latency |
|---|---|---|---|---|---|---|
| German Credit | Jev | 0 | 0.467 | – | 0.022 | 0.15 s |
| | Qwen3.6-27B | 0 | 0.457 | – | 0.683 (0.642) | 0.95 s (2.5/s) |
| | XGBoost | 600 | 0.657 | 0.0097 | <0.0001 | 1.3 $\mu$s |
| | BERT | 600 | 0.473 | 0.148 | 0.0003 | 0.76 ms |
| Breast Cancer Wisconsin | Jev | 0 | 0.343 | – | 0.030 | 0.14 s |
| | Qwen3.6-27B | 0 | 0.783 | – | 1.34 (1.26) | 1.10 s (1.3/s) |
| | XGBoost | 341 | 0.945 | 0.010 | <0.0001 | 3.1 $\mu$s |
| | BERT | 341 | 0.716 | 0.129 | 0.0004 | 1.26 ms |
| SST-2 | Jev | 0 | 0.960 | – | 0.015 | 0.15 s |
| | Qwen3.6-27B | 0 | 0.956 | – | 0.352 (0.331) | 0.92 s (4.8/s) |
| | XGBoost | 65,349 | 0.791 | 0.024 | <0.0001 | 0.01 ms |
| | BERT | 65,349 | 0.916 | 0.457 | <0.0001 | 0.19 ms |
| AG News | Jev | 0 | 0.878 | – | 0.018 | 0.15 s |
| | Qwen3.6-27B | 0 | 0.877 | – | 0.586 (0.551) | 0.93 s (2.9/s) |
| | XGBoost | 115,000 | 0.914 | 0.157 | <0.0001 | 0.06 ms |
| | BERT | 115,000 | 0.942 | 1.09 | 0.0001 | 0.31 ms |

tabular model by a wide margin (accuracy 0.741 and 0.949); BERT, reading serialized rows, stays near the majority class on German Credit (macro-F1 0.473).

Because a miss costs more than a false alarm, the error counts matter more than accuracy here (Table 3). Jev's bias is cheap under this cost: at the default decision it misses only 126 of 600 bad applicants and 3 of 420 malignant masses, at the price of 945 and 675 false alarms, and its cost on German Credit (1,575) is the lowest of the four models. That is not skill: calling every case positive costs 1,400 and 720, less than Jev on German Credit and about the same on Breast Cancer Wisconsin. With a cost-selected threshold XGBoost is cheapest on both (1,132 and 168), and on German Credit neither zero-shot model improves on calling every applicant bad (1,395 and 1,426 against 1,400).

## 6 Training labels, cost and latency

A zero-shot model is paid per prediction and needs no labels; a supervised one needs labels and a training run and then predicts almost for free. Figure 1 puts the four models on one axis and Table 4 adds what each costs and how long a prediction takes. Every cost is computed from recorded quantities: Jev's token counts at its published price, and the billable seconds of each cloud job at the instance's list price.

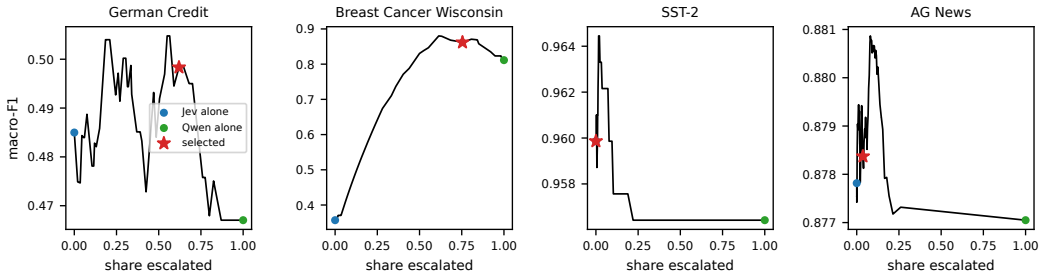



Figure 2: Macro-F1 of the ensemble against the share of evaluation items escalated to Qwen (first split repeat for the tabular datasets). The star marks the validation-selected threshold.

Table 5: Escalation of low-confidence Jev predictions to Qwen3.6-27B: macro-F1 of each model alone and of the ensemble at the validation-selected threshold $t$, the share of items escalated, the cost per 1,000 predictions, and the Jev errors at its highest reported confidence, which no threshold can reach without escalating every item at that confidence.

| Dataset | Jev | Qwen | $t$ | Share | Ensemble | USD/1,000 | Unreachable |
|---|---|---|---|---|---|---|---|
| German Credit | 0.467 $\pm$0.019 | 0.457 $\pm$0.025 | 0.88 | 0.58 | 0.478 $\pm$0.029 | 0.421 | 5 of 103 |
| Breast Cancer Wisconsin | 0.343 $\pm$0.019 | 0.783 $\pm$0.029 | 0.85 | 0.69 | 0.856 $\pm$0.027 | 0.950 | 0 of 66 |
| SST-2 | 0.960 | 0.956 | 0.00 | 0.00 | 0.960 | 0.015 | 5 of 35 |
| AG News | 0.878 | 0.877 | 0.65 | 0.04 | 0.878 | 0.040 | 307 of 923 |

With enough labels a fine-tuned encoder overtakes both zero-shot models on topic classification: BERT reaches macro-F1 0.942 on AG News against 0.878 for Jev, and first matches Jev at 5,752 training labels (XGBoost: 28,752). On SST-2 neither supervised model matches the zero-shot ones at any size tried: BERT reaches 0.916 with all 65,349 labels, against 0.960 for Jev. On the tabular datasets XGBoost needs only its smallest training set to exceed both zero-shot models.

A Jev prediction cost USD 0.015 to USD 0.030 per 1,000, depending on prompt length, and took 0.15 s. Self-hosting Qwen on `ml.g6.12xlarge` cost USD 0.352 to USD 1.342 per 1,000, which is 24 to 44 times Jev's price, at 0.92 to 1.10 s per single item. The supervised models are cheaper per prediction by orders of magnitude once trained; their cost is the training run (at most USD 1.09 for BERT) and, above all, the labels. Letting Qwen reason before answering raised its macro-F1 on every dataset on a 50-item sample (for example 0.772 to 0.890 on Breast Cancer Wisconsin) at many times the compute (Table 15). Appendix D gives the accuracy curves (Figure 5) and every point of both curves (Table 14).

## 7 Jev with Qwen in the loop

A class distribution also says when a second opinion is worth paying for. We keep Jev's prediction when its maximum class probability is at least a threshold $t$ and take Qwen's otherwise. The reported point uses the $t$ that maximizes macro-F1 on validation items only, with ties broken towards fewer escalations; everything is computed from saved predictions (Figure 2; Appendix E lists the validation items).

The outcome depends on what Jev's confidence knows (Table 5). On Breast Cancer Wisconsin the ensemble beats both of its members: macro-F1 0.856 against 0.343 for Jev and 0.783 for Qwen, escalating 0.69 of the items. Jev's confident predictions there are mostly right, and the ones it is unsure of are better left to Qwen. On German Credit, where neither model ranks the rows, escalating 0.58 of them moves macro-F1 only from 0.467 to 0.478. On SST-2 the selected threshold escalates nothing. On AG News 0.04 of the items are escalated and macro-F1 stays at 0.878: the two models are equally accurate, and 307 of Jev's 923 errors carry its highest reported confidence, shared by 0.74 of all items. Escalation is paid for at Qwen's price: USD 0.950 per 1,000 predictions on Breast Cancer Wisconsin against USD 0.030 for Jev alone.

Table 6: Mean decision cost at the default and the analytic threshold, the validation-selected threshold $t$ with its cost, and accuracy before and after selection. Tabular values are means over 10 split repeats. All models and all metrics are in Table 16, and further sweeps in Figure 6.

| | | Cost at fixed $t$ | | Validation-selected | | Accuracy | |
|---|---|---|---|---|---|---|---|
| Dataset | Model | default | analytic | $t$ | cost | default | selected |
| German Credit | Jev | 0.787 | 0.686 | 0.17 | 0.698 | 0.465 | 0.350 |
| | XGBoost | 0.963 | 0.544 | 0.17 | 0.566 | 0.741 | 0.588 |
| Breast Cancer Wisconsin | Jev | 0.605 | 0.630 | 0.77 | 0.395 | 0.405 | 0.735 |
| | XGBoost | 0.170 | 0.124 | 0.27 | 0.147 | 0.949 | 0.937 |
| SST-2 | Jev | 0.040 | 0.040 | 0.36 | 0.033 | 0.960 | 0.967 |
| | XGBoost | 0.209 | 0.209 | 0.55 | 0.205 | 0.791 | 0.795 |

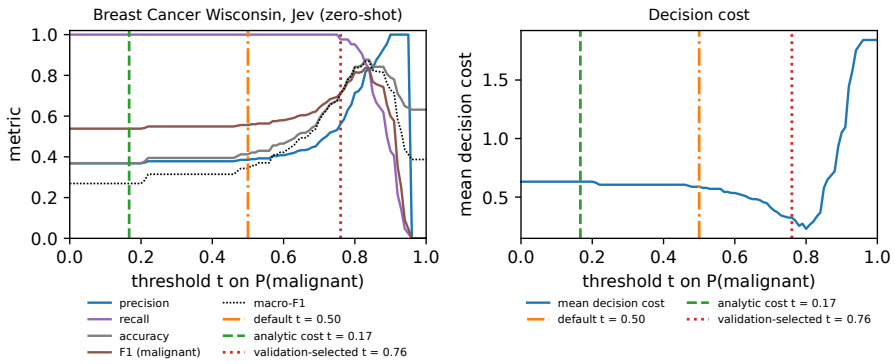



Figure 3: Jev on Breast Cancer Wisconsin Diagnostic, first split repeat. Left: precision, recall, accuracy and positive-class F1 against the threshold $t$ (macro-F1 dashed). Right: mean decision cost against $t$. The three thresholds are marked in both panels.

## 8 Threshold selection

With class probabilities a decision can be tuned after the fact: predict the positive class when $\mu_x(\{\text{positive}\}) \geq t$. We compare the default $t$, the analytic cost threshold $C_{\text{FP}}/(C_{\text{FP}} + C_{\text{FN}})$, which is optimal under calibration [5], and the threshold that minimizes mean decision cost on validation examples. German Credit has a canonical cost matrix (5:1); for Breast Cancer Wisconsin we assume 5:1 for a missed malignant case and for SST-2 symmetric costs.

Breast Cancer Wisconsin shows what the probabilities are worth when the default decision is not (Table 6, Figure 3). Jev's probability of *malignant* ranks the rows well but sits too high. The analytic threshold assumes calibration and moves the wrong way. The validation-selected threshold moves to $t = 0.77$: accuracy rises from 0.405 to 0.735 and the cost falls from 0.605 to 0.395, using 114 labels and no training. It does not close the gap to XGBoost (0.147). On German Credit no threshold helps Jev much, because there is little ranking to exploit; for the calibrated XGBoost the analytic threshold works as theory says, cutting its cost from 0.963 to 0.544. On SST-2, where Jev is already accurate and calibrated, selection changes little. A model that returns only a label is fixed at one operating point; a distribution lets about a hundred labels place the decision.

## 9 Estimating accuracy from few labels

Labels are often the bottleneck for evaluating a deployed classifier. With class probabilities the predicted uncertainty $u_i = 1 - p_{\max,i}$ can guide which items to label. We compare uniform sampling with the confidence-guided proposal $q_i = (1-\varepsilon)\, u_i^\alpha / \sum_j u_j^\alpha + \varepsilon/N$ with $\varepsilon = 0.1$ and $\alpha = 1$ [11, 18], estimating the error rate with the Hansen–Hurwitz estimator [7], over 1,000 seeded trials per label budget; the sampler never sees a label it did not draw.

Table 7: RMSE of the accuracy estimate from $n$ labels, for uniform and confidence-guided sampling, and the ratio of their variances (uniform over guided; above 1 favours guided sampling), over 1,000 trials.

| | | | $n = 25$ | | | $n = 100$ | | | $n = 400$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Dataset | Pool | Accuracy | uniform | guided | ratio | uniform | guided | ratio | uniform | guided | ratio |
| German Credit | 1,000 | 0.471 | 0.097 | 0.212 | 0.21 | 0.049 | 0.105 | 0.22 | 0.025 | 0.052 | 0.23 |
| Breast Cancer Wisconsin | 569 | 0.427 | 0.101 | 0.090 | 1.27 | 0.048 | 0.045 | 1.14 | 0.024 | 0.021 | 1.32 |
| SST-2 | 872 | 0.960 | 0.039 | 0.057 | 0.46 | 0.020 | 0.029 | 0.47 | 0.010 | 0.014 | 0.48 |
| AG News | 7,600 | 0.879 | 0.067 | 0.135 | 0.24 | 0.032 | 0.069 | 0.22 | 0.017 | 0.032 | 0.28 |

Table 8: Explanations of Jev's predictions: mean drop in the predicted-class probability and share of label flips after deleting the top three attributed units, against three random units; Kendall's $\tau$ and top-3 overlap between KernelSHAP and LIME; and Jev requests used.

| | | | Probability drop, $k = 3$ | | | Label flips, $k = 3$ | | Agreement | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Dataset | Unit | Units | SHAP | LIME | random | SHAP | random | $\tau$ | top-3 | Jev requests |
| German Credit | feature | 20 | 0.577 | 0.603 | 0.008 | 0.85 | 0.06 | 0.71 | 0.80 | 7,164 |
| Breast Cancer Wisconsin | feature | 30 | 0.065 | 0.155 | 0.010 | 0.05 | 0.02 | 0.37 | 0.35 | 7,969 |
| SST-2 | word | 18 | 0.401 | 0.419 | 0.096 | 0.40 | 0.09 | 0.51 | 0.70 | 6,666 |
| AG News | word | 34 | 0.075 | 0.091 | 0.003 | 0.10 | 0.00 | 0.19 | 0.27 | 7,919 |

The estimator is unbiased in practice (largest absolute bias 0.005), but the guided proposal reduces variance on only 1 of the 4 datasets (Table 7). At 400 labels the ratio of uniform to guided variance is 1.32 on Breast Cancer Wisconsin and 0.23, 0.48 and 0.28 on German Credit, SST-2 and AG News. The distribution itself shows why. Jev rounds probabilities to two decimals, so 5,621 of 7,600 AG News items have a reported uncertainty of exactly zero; they receive only the minimum sampling probability yet contain 307 of the 923 errors, each weighted by $1/\varepsilon$. And for a 0/1 loss a proposal proportional to the expected loss is not variance-optimal even under perfect calibration. The check that reveals this needs only a small labelled sample (Appendix G, Figure 7).

## 10 Perturbation-based explanations

Model-agnostic attribution needs only a real-valued function of the input, and with a class distribution it is immediate: $f(z) = \mu_{x(z)}(\{c^\star\})$, Jev's probability of the class it predicts on the full input when only the units in the coalition $z$ are kept (a feature line for tabular rows, a word for text). We ran KernelSHAP [13] and LIME [17] on 20 seeded examples per dataset, on live Jev, and tested faithfulness by deleting the top-attributed units and re-querying.

Both methods run unmodified, and how useful they are depends on the dataset (Table 8). On German Credit and SST-2 the explanations are faithful: deleting the three top KernelSHAP units lowers the predicted-class probability by 0.577 and 0.401, against 0.008 and 0.096 for three random units. On Breast Cancer Wisconsin and AG News they are weak (0.065 and 0.075): with 30 correlated measurements, or 34 words, no three units carry the decision. The base value is informative in itself: with every feature removed, Jev still gives *bad* a probability of 0.69 on German Credit, the bias that Section 5 measures. These explanations cost 29,718 requests; Appendix H shows the deletion curves (Figure 8) and example attributions (Figure 9).

## 11 Limitations

The comparison is unequal by design: two models use no training labels and two use up to 115,000. Each dataset has one frozen template, never tuned; a public result on Breast Cancer Wisconsin with a different serialization is much higher than ours, so our tabular numbers describe this template, not Jev's best case. Qwen's distribution comes from scoring option names, one of several ways to turn a generative model into a classifier, and carries numerical noise at the second decimal; its cost and latency describe one deployment. No model is recalibrated. The tabular datasets are small and their repeated splits overlap. Only German Credit has a canonical cost matrix. Thinking was measured on 50 items per dataset, and explanations on 20. All four datasets are widely circulated and the models

may have seen them. A fifth dataset was run and removed before writing; its costs are reported (Appendix B).

## 12 Conclusion

A generative language model defines a measure on token strings; classes are reached only through a parsing map that leaks probability, and the class distribution can only be estimated. A model that returns a probability for every option defines the measure on $(C, 2^C)$ directly. On four datasets that was enough to apply calibration analysis, threshold selection, escalation, importance-sampled evaluation and perturbation-based explanations to Jev with standard tools, and to score it with the same code as supervised classifiers; obtaining the same object from a generative model took a scoring procedure of its own and far more computation per prediction. What the toolbox found is mixed, and that is its value: accurate, usable probabilities on text; a bias towards the adverse class on tables that a threshold or a second model repairs in one case and not in the other; and guided sampling and escalation that pay off only where the reported confidence is informative. The first thing the classical toolbox reports about a class distribution is where it can be trusted.

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# A Prior public evaluations of Jev

Table 9 sets the public results next to ours. Every prior number in it was read from its source on 2026-09-30 and is stored with the quoted text it came from; none was typed into this paper. The protocols differ: each study uses its own template; the repository results are for samples, with ECE over 10 bins; and the tabular benchmark uses a single 114-row split and a different serialization, which is the likely reason its accuracy on Breast Cancer Wisconsin Diagnostic (0.842) is far above ours (0.411).

Table 9: Prior public evaluations of zero-shot Jev on the datasets used here, next to our results at the default decision. Accuracy is at the argmax label.

| | | Prior public evaluation | | | This paper (Jev, zero-shot) | | |
|---|---|---|---|---|---|---|---|
| Dataset | Source | $n$ | Acc. | ECE | $n$ | Acc. | ECE |
| SST-2 | Deußer et al. [3] | 872 | 0.964 | 0.015 | 872 | 0.960 | 0.020 |
| AG News | Deußer et al. [3] | 7,600 | 0.885 | 0.077 | 7,600 | 0.879 | 0.082 |
| SST-2 | jevbench [14] | 500 | 0.954 | 0.026 | 872 | 0.960 | 0.020 |
| AG News | jevbench [14] | 500 | 0.843 | 0.112 | 7,600 | 0.879 | 0.082 |
| Breast Cancer Wisconsin Diagnostic | jev-classification-benchmark [20] | 114 | 0.842 | – | 114 | 0.411 | 0.374 |
| German Credit | none found | – | – | – | 200 | 0.467 | 0.364 |

# B Protocol, label budgets and cost

**Models.** Jev is queried through the `typesafe-sdk` Python client (version 0.7.2) with one `Choice` question per example; a missing tabular value is written as `unknown`, and text is sent after collapsing whitespace. The text templates were tried once on 20 training-split examples and frozen unchanged; the tabular templates were frozen without being tried on data rows, because every row can be an evaluation row in some repeat. XGBoost is trained with a fixed grid of 4 configurations (tree depth and learning rate), early stopping on the validation part (at most 500 rounds, patience 30) and, for text, TF-IDF weights over unigrams and bigrams (at most 50,000 features, 16 histogram bins). BERT is `google-bert/bert-base-uncased` (revision `86b5e0934494`); learning rates 2e-05 and 5e-05 are each trained for up to 4 epochs (batch size 32) and the learning rate and epoch with the lowest validation log-loss are kept, on the first split repeat for the tabular datasets. Inputs are truncated at 512 tokens (tabular), 256 (AG News) or 128 (SST-2). Qwen is `Qwen/Qwen3.6-27B` (revision `6a9e13bd6fc8`) in bf16. Its scores are computed with `transformers` 5.16.1 on `ml.g6.12xlarge`: the instruction and option list once per dataset, the rest of the prompt once per item, and every option on a copy of that state, after a check on one item per dataset that this equals a full recomputation. With thinking enabled the reasoning (greedy, at most 2,048 tokens) is generated with vLLM [12] 0.29.0 and the options are scored after it. All supervised training and all Qwen inference ran as managed cloud training jobs (containers `sagemaker-xgboost:3.0-5` and `huggingface-pytorch-training:2.9.0-transformers5.3.0-gpu-py312-cu130-ubuntu22.04` for training).

Table 10: Labels used by each model, per split repeat. "Early stopping, selection" counts the validation labels used for early stopping and hyperparameter selection. Threshold selection applies to the binary datasets and reuses validation labels. The escalation threshold is selected on validation items predicted by both zero-shot models.

| Dataset | Model | Training | Early stopping, selection | Calibration | Threshold selection | Escalation threshold |
|---|---|---|---|---|---|---|
| German Credit | Jev, zero-shot | 0 | 0 | 0 | 200 | 200 |
| | Qwen3.6-27B, zero-shot | 0 | 0 | 0 | 200 | 200 |
| | XGBoost, supervised | 600 | 200 | 0 | 200 | – |
| | BERT, supervised | 600 | 200 | 0 | 200 | – |
| Breast Cancer Wisconsin | Jev, zero-shot | 0 | 0 | 0 | 114 | 114 |
| | Qwen3.6-27B, zero-shot | 0 | 0 | 0 | 114 | 114 |
| | XGBoost, supervised | 341 | 114 | 0 | 114 | – |
| | BERT, supervised | 341 | 114 | 0 | 114 | – |
| SST-2 | Jev, zero-shot | 0 | 0 | 0 | 500 | 500 |
| | Qwen3.6-27B, zero-shot | 0 | 0 | 0 | 500 | 500 |
| | XGBoost, supervised | 65,349 | 2,000 | 0 | 2,000 | – |
| | BERT, supervised | 65,349 | 2,000 | 0 | 2,000 | – |
| AG News | Jev, zero-shot | 0 | 0 | 0 | 0 | 500 |
| | Qwen3.6-27B, zero-shot | 0 | 0 | 0 | 0 | 500 |
| | XGBoost, supervised | 115,000 | 5,000 | 0 | 0 | – |
| | BERT, supervised | 115,000 | 5,000 | 0 | 0 | – |

**Budget and cost.** The study was capped at 60,000 Jev inferences and used 50,000 (13,620 for predictions, 35,317 for explanations, 1,000 for escalation validation items, 60 for template checks, 3 for connectivity checks). Every request is counted in a persistent ledger before it is sent, and client retries are disabled; 0 requests failed. Token counts exist for 49,999 of the requests: 39.0 million input tokens, USD 1.64 at USD 0.042 per million (Table 11). All cloud jobs together cost about USD 25.67 at list prices, under a limit of USD 75: USD 0.67 for XGBoost, USD 1.97 for BERT (1.57 GPU hours) and USD 23.02 for Qwen (3.75 GPU hours), counting stopped, failed and repeated executions. XGBoost was run a second time on every dataset to record its inference time, and its predictions were identical to the first ones. Costs are computed from recorded token counts and from job records at list prices, not from invoices.

**A dataset that was run and removed.** A fifth dataset, Banking77, was run and then removed from the study, because the four-model comparison could not be completed on it within the compute limit. Everything spent on it remains in the totals above: 9,198 Jev requests (21.1 million input tokens, USD 0.89), USD 1.05 of cloud jobs dedicated to it (5 executions), and a part of the USD 12.77 spent on 9 exploratory Qwen executions that also covered other datasets and cannot be split. No result for it is reported.

**Societal impact.** Two of the datasets stand for decisions that affect people: credit and medical diagnosis. Our results are a caution for such uses. A zero-shot model's default decision was biased towards the adverse class on both, on German Credit even with no evidence in the input, and nothing in its text output would have revealed it. The positive side is that a class distribution makes the bias measurable with standard tools and, where the ranking is sound, repairable with a small labelled sample. The results describe four public benchmarks and one template per dataset and should not be read as evidence that any of these models is fit or unfit for a deployment.

**Uncertainty.** For the held-out datasets we report 95% bootstrap intervals (1,000 resamples, fixed seed). For the tabular datasets we report the mean and standard deviation over the 10 repeats; their test parts overlap across repeats, so the standard deviation describes split variation, not an independent-sample standard error.

Table 11: Jev requests by stage. Input tokens are read from the logged responses and, for the 99 cached responses that predate the log, from the cache; 1 request has no recorded token count. Output is not billed. The last row is the part of the total that was spent on the removed dataset.

| Stage | Requests | In request log | Input tokens (M) | Cost (USD) |
|---|---|---|---|---|
| smoke | 3 | 2 | 0.00 | 0.000 |
| template | 60 | 60 | 0.06 | 0.003 |
| predict | 13,620 | 13,521 | 11.80 | 0.496 |
| explain | 35,317 | 35,317 | 25.81 | 1.084 |
| escalation_val | 1,000 | 1,000 | 1.37 | 0.057 |
| all stages | 50,000 | 49,900 | 39.04 | 1.640 |
| of which Banking77 (removed) | 9,198 | 9,198 | 21.10 | 0.886 |

# C Full metrics and calibration

Table 12: Classification metrics on identical evaluation examples. Brackets: 95% bootstrap intervals (held-out datasets). ±: standard deviation over 10 split repeats (tabular datasets).

| Dataset | Model | Accuracy | Macro-F1 | ROC-AUC | PR-AUC |
|---|---|---|---|---|---|
| German Credit | Jev, zero-shot | 0.467 ±0.019 | 0.467 ±0.019 | 0.537 ±0.030 | 0.316 ±0.023 |
| | Qwen3.6-27B, zero-shot | 0.544 ±0.023 | 0.457 ±0.025 | 0.496 ±0.035 | 0.304 ±0.028 |
| | XGBoost, supervised | 0.741 ±0.031 | 0.657 ±0.041 | 0.768 ±0.031 | 0.574 ±0.046 |
| | BERT, supervised | 0.705 ±0.008 | 0.473 ±0.083 | 0.683 ±0.069 | 0.467 ±0.071 |
| Breast Cancer Wisconsin | Jev, zero-shot | 0.411 ±0.014 | 0.343 ±0.019 | 0.917 ±0.021 | 0.878 ±0.032 |
| | Qwen3.6-27B, zero-shot | 0.825 ±0.020 | 0.783 ±0.029 | 0.969 ±0.012 | 0.957 ±0.012 |
| | XGBoost, supervised | 0.949 ±0.021 | 0.945 ±0.023 | 0.985 ±0.015 | 0.982 ±0.015 |
| | BERT, supervised | 0.778 ±0.092 | 0.716 ±0.182 | 0.853 ±0.057 | 0.829 ±0.044 |
| SST-2 | Jev, zero-shot | 0.960 [0.948, 0.971] | 0.960 [0.948, 0.971] | 0.987 [0.979, 0.993] | 0.986 [0.976, 0.993] |
| | Qwen3.6-27B, zero-shot | 0.956 [0.943, 0.968] | 0.956 [0.943, 0.968] | 0.989 [0.984, 0.994] | 0.990 [0.984, 0.995] |
| | XGBoost, supervised | 0.791 [0.763, 0.819] | 0.791 [0.762, 0.818] | 0.873 [0.848, 0.896] | 0.879 [0.849, 0.905] |
| | BERT, supervised | 0.916 [0.896, 0.935] | 0.916 [0.896, 0.935] | 0.976 [0.968, 0.983] | 0.979 [0.971, 0.986] |
| AG News | Jev, zero-shot | 0.879 [0.871, 0.886] | 0.878 [0.870, 0.885] | 0.961 [0.958, 0.964] | 0.901 [0.893, 0.910] |
| | Qwen3.6-27B, zero-shot | 0.877 [0.870, 0.885] | 0.877 [0.870, 0.884] | 0.962 [0.958, 0.965] | 0.909 [0.900, 0.917] |
| | XGBoost, supervised | 0.914 [0.908, 0.921] | 0.914 [0.908, 0.921] | 0.985 [0.983, 0.986] | 0.960 [0.955, 0.965] |
| | BERT, supervised | 0.942 [0.936, 0.947] | 0.942 [0.937, 0.947] | 0.992 [0.991, 0.993] | 0.979 [0.976, 0.982] |

Table 13: Calibration: expected calibration error (15 equal-width bins), Brier score, log-loss, and mean confidence (mean of the maximum class probability). A reported probability of exactly zero for the true class is clipped at $10^{-12}$ in the log-loss.

| Dataset | Model | ECE | Brier | Log-loss | Mean confidence |
|---|---|---|---|---|---|
| German Credit | Jev, zero-shot | 0.364 ±0.030 | 0.833 ±0.041 | 2.136 ±0.291 | 0.814 |
| | Qwen3.6-27B, zero-shot | 0.136 ±0.015 | 0.499 ±0.017 | 0.694 ±0.019 | 0.669 |
| | XGBoost, supervised | 0.069 ±0.017 | 0.346 ±0.022 | 0.516 ±0.027 | 0.757 |
| | BERT, supervised | 0.060 ±0.034 | 0.390 ±0.024 | 0.574 ±0.029 | 0.706 |
| Breast Cancer Wisconsin | Jev, zero-shot | 0.374 ±0.013 | 0.655 ±0.014 | 0.865 ±0.020 | 0.773 |
| | Qwen3.6-27B, zero-shot | 0.072 ±0.026 | 0.223 ±0.012 | 0.341 ±0.015 | 0.857 |
| | XGBoost, supervised | 0.036 ±0.010 | 0.074 ±0.032 | 0.139 ±0.064 | 0.966 |
| | BERT, supervised | 0.115 ±0.052 | 0.312 ±0.062 | 0.486 ±0.075 | 0.747 |
| SST-2 | Jev, zero-shot | 0.020 [0.014, 0.034] | 0.064 [0.047, 0.082] | 0.255 [0.138, 0.397] | 0.967 |
| | Qwen3.6-27B, zero-shot | 0.023 [0.017, 0.038] | 0.073 [0.054, 0.093] | 0.159 [0.112, 0.214] | 0.974 |
| | XGBoost, supervised | 0.026 [0.022, 0.061] | 0.290 [0.261, 0.320] | 0.451 [0.411, 0.493] | 0.799 |
| | BERT, supervised | 0.054 [0.040, 0.075] | 0.138 [0.109, 0.170] | 0.259 [0.204, 0.321] | 0.966 |
| AG News | Jev, zero-shot | 0.082 [0.076, 0.090] | 0.204 [0.192, 0.216] | 1.464 [1.337, 1.591] | 0.960 |
| | Qwen3.6-27B, zero-shot | 0.099 [0.093, 0.107] | 0.219 [0.207, 0.233] | 0.925 [0.857, 0.997] | 0.976 |
| | XGBoost, supervised | 0.032 [0.027, 0.038] | 0.135 [0.127, 0.143] | 0.270 [0.256, 0.286] | 0.887 |
| | BERT, supervised | 0.018 [0.014, 0.023] | 0.091 [0.084, 0.100] | 0.179 [0.165, 0.197] | 0.958 |

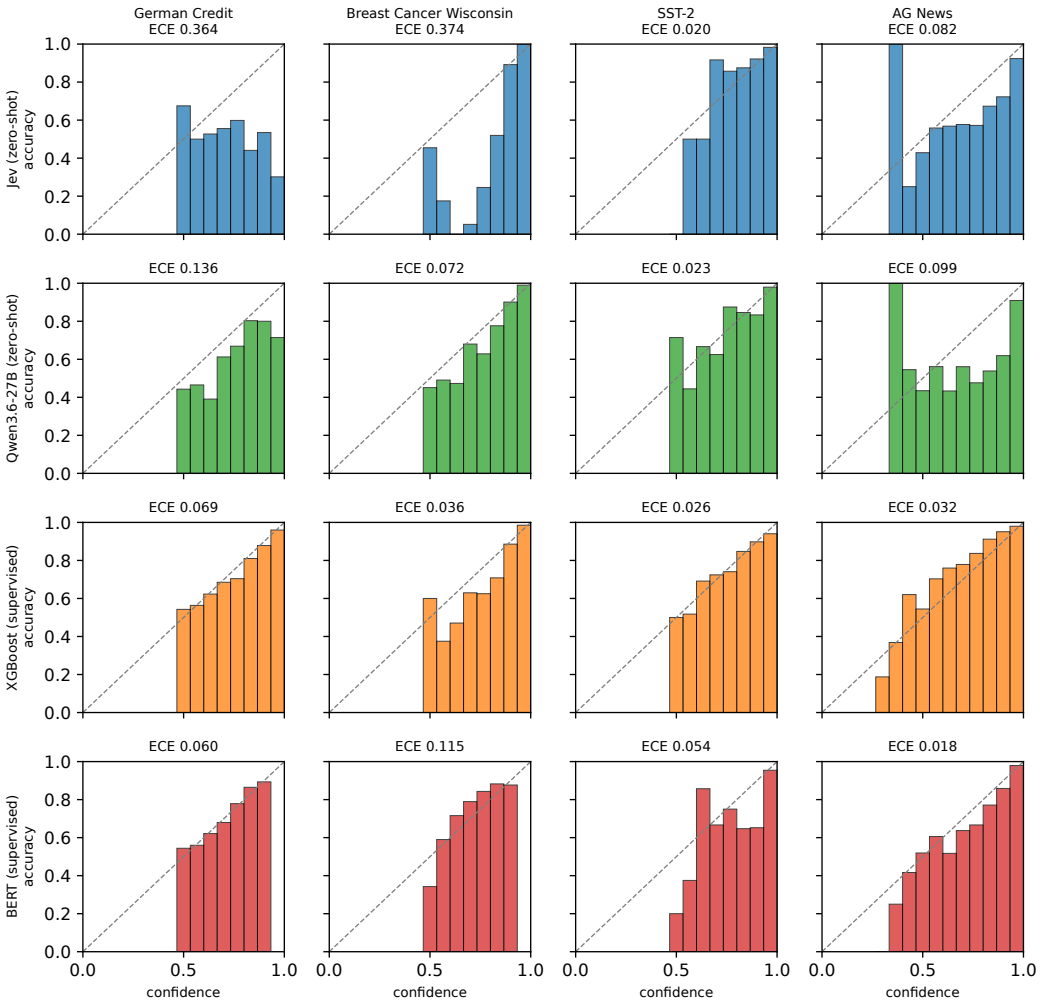



Figure 4: Reliability diagrams: accuracy against the maximum class probability, in 15 equal-width bins. Rows, top to bottom: Jev and Qwen3.6-27B (zero-shot), XGBoost and BERT (supervised). Tabular datasets pool the test parts of all split repeats.

## D Learning curves and thinking

On the text datasets, XGBoost sees only TF-IDF features and reaches 0.791 on SST-2 and 0.914 on AG News (zero-shot Jev: 0.960 and 0.879). The learning curves (Figure 5) put the comparison in terms of labels: on SST-2 it does not reach zero-shot Jev even with all 65,349 training labels; on AG News it first reaches zero-shot Jev at 28,752 training labels.

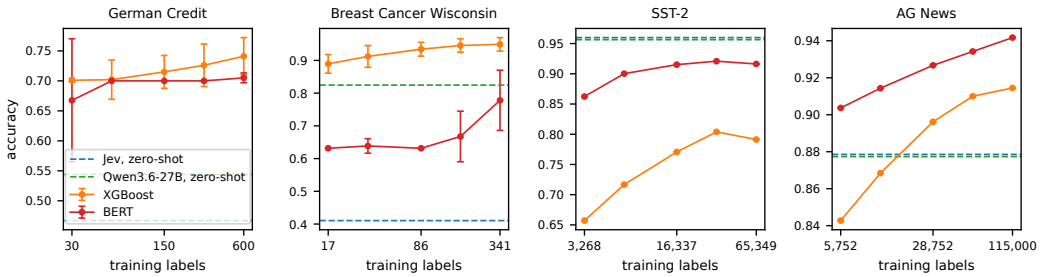



Figure 5: Accuracy of XGBoost and BERT on the evaluation examples against the number of training labels, with the zero-shot models as horizontal references.

Table 14: XGBoost and BERT on the evaluation examples when trained on nested, stratified fractions of the training part with their selected configuration (mean over split repeats for the tabular datasets). The first row of each dataset gives the zero-shot models, at zero training labels, as accuracy / macro-F1.

| | | XGBoost, supervised | | BERT, supervised | |
|---|---|---|---|---|---|
| Dataset | Training labels | Accuracy | Macro-F1 | Accuracy | Macro-F1 |
| German Credit | 0 | Jev 0.467 / 0.467; Qwen3.6-27B 0.544 / 0.457 | | | |
| | 30 | 0.701 | 0.441 | 0.667 | 0.407 |
| | 60 | 0.702 | 0.531 | 0.700 | 0.412 |
| | 150 | 0.715 | 0.574 | 0.700 | 0.412 |
| | 300 | 0.726 | 0.624 | 0.700 | 0.412 |
| | 600 | 0.741 | 0.657 | 0.705 | 0.473 |
| Breast Cancer Wisconsin | 0 | Jev 0.411 / 0.343; Qwen3.6-27B 0.825 / 0.783 | | | |
| | 17 | 0.889 | 0.879 | 0.632 | 0.387 |
| | 34 | 0.912 | 0.904 | 0.639 | 0.408 |
| | 86 | 0.934 | 0.929 | 0.632 | 0.387 |
| | 171 | 0.946 | 0.941 | 0.668 | 0.463 |
| | 341 | 0.949 | 0.945 | 0.778 | 0.716 |
| SST-2 | 0 | Jev 0.960 / 0.960; Qwen3.6-27B 0.956 / 0.956 | | | |
| | 3,268 | 0.657 | 0.654 | 0.862 | 0.861 |
| | 6,535 | 0.717 | 0.713 | 0.900 | 0.900 |
| | 16,337 | 0.771 | 0.769 | 0.915 | 0.915 |
| | 32,674 | 0.804 | 0.804 | 0.921 | 0.921 |
| | 65,349 | 0.791 | 0.791 | 0.916 | 0.916 |
| AG News | 0 | Jev 0.879 / 0.878; Qwen3.6-27B 0.877 / 0.877 | | | |
| | 5,752 | 0.843 | 0.842 | 0.904 | 0.904 |
| | 11,500 | 0.868 | 0.868 | 0.914 | 0.914 |
| | 28,752 | 0.896 | 0.896 | 0.927 | 0.927 |
| | 57,500 | 0.910 | 0.910 | 0.934 | 0.934 |
| | 115,000 | 0.914 | 0.914 | 0.942 | 0.942 |

**Thinking.** On the 50 items per dataset where Qwen also reasoned before answering, its macro-F1 rose on every dataset: from 0.772 to 0.890 on Breast Cancer Wisconsin Diagnostic, 0.435 to 0.531 on German Credit, 0.918 to 0.959 on SST-2 and 0.854 to 0.893 on AG News. These are small samples, and the reasoning is long (1,500 tokens on average on Breast Cancer Wisconsin Diagnostic, with 0.40 of German Credit items reaching the limit). Reasoning on every item was projected at USD 49, which does not fit our spending limit with a safety margin, so it was not run.

Table 15: Macro-F1 on the thinking subsample. Every model is scored on the same items. "Thinking tokens" is the mean length of the reasoning and "Truncated" the share of items whose reasoning reached the limit of 2,048 tokens.

| Dataset | $n$ | Jev | Qwen, off | Qwen, on | XGBoost | BERT | Thinking tokens | Truncated |
|---|---|---|---|---|---|---|---|---|
| German Credit | 50 | 0.440 | 0.435 | 0.531 | 0.688 | 0.479 | 1,765 | 0.40 |
| Breast Cancer Wisconsin | 50 | 0.319 | 0.772 | 0.890 | 0.934 | 0.792 | 1,500 | 0.12 |
| SST-2 | 50 | 0.939 | 0.918 | 0.959 | 0.760 | 0.960 | 376 | 0.00 |
| AG News | 50 | 0.873 | 0.854 | 0.893 | 0.884 | 0.938 | 522 | 0.00 |

# E Escalation: validation items

The validation items for the escalation threshold are each repeat's validation part for the tabular datasets, the 500 threshold-selection items for SST-2, and 500 seeded validation items for AG News that both models predicted for this purpose.

# F Threshold selection in full

Jev's threshold is selected on 200, 114 and 500 validation labels for German Credit, Breast Cancer Wisconsin Diagnostic and SST-2; on the tabular datasets this is repeated for each of the 10 splits. On Breast Cancer Wisconsin Diagnostic, at the default threshold Jev calls 0.95 of the rows malignant, for a mean cost of 0.605; the analytic threshold (0.17) moves the wrong way, to 0.630; the validation-selected threshold reaches 0.395, with recall 0.912 and precision 0.605 on malignant cases. On German Credit the cost is 0.787 at the default, 0.686 at the analytic and 0.698 at the validation-selected threshold, and the lower cost is bought by calling nearly every applicant bad (recall 0.960, precision 0.311). On SST-2 the selected threshold ($t = 0.36$) changes the cost only from 0.040 to 0.033.

Table 16: Decisions at three thresholds on P(positive) for every model. Positive classes: bad (German Credit), malignant (Breast Cancer Wisconsin), positive (SST-2). Tabular values are means ± standard deviations over 10 split repeats, and $t$ is the mean threshold. Costs: 5:1 (canonical), 5:1 (illustrative), 1:1 (illustrative).

| Dataset | Model | Threshold | $t$ | Precision | Recall | F1 (pos.) | Accuracy | Mean cost |
|---|---|---|---|---|---|---|---|---|
| German Credit | Jev, zero-shot | default | 0.50 | 0.334 ±0.012 | 0.790 ±0.030 | 0.470 ±0.016 | 0.465 ±0.020 | 0.787 ±0.048 |
| | | analytic | 0.17 | 0.313 ±0.007 | 0.968 ±0.018 | 0.473 ±0.010 | 0.351 ±0.016 | 0.686 ±0.034 |
| | | validation-selected | 0.17 | 0.311 ±0.008 | 0.960 ±0.027 | 0.470 ±0.011 | 0.350 ±0.017 | 0.698 ±0.042 |
| | Qwen3.6-27B, zero-shot | default | 0.50 | 0.244 ±0.037 | 0.252 ±0.043 | 0.248 ±0.039 | 0.542 ±0.023 | 1.356 ±0.068 |
| | | analytic | 0.17 | 0.314 ±0.009 | 0.955 ±0.026 | 0.472 ±0.012 | 0.360 ±0.020 | 0.694 ±0.043 |
| | | validation-selected | 0.14 | 0.306 ±0.008 | 0.960 ±0.039 | 0.464 ±0.012 | 0.335 ±0.024 | 0.713 ±0.046 |
| | XGBoost, supervised | default | 0.50 | 0.604 ±0.084 | 0.413 ±0.071 | 0.488 ±0.064 | 0.741 ±0.031 | 0.963 ±0.106 |
| | | analytic | 0.17 | 0.419 ±0.017 | 0.883 ±0.047 | 0.568 ±0.021 | 0.597 ±0.026 | 0.544 ±0.062 |
| | | validation-selected | 0.17 | 0.415 ±0.031 | 0.872 ±0.091 | 0.560 ±0.031 | 0.588 ±0.057 | 0.566 ±0.096 |
| | BERT, supervised | default | 0.50 | 0.231 ±0.306 | 0.093 ±0.137 | 0.125 ±0.173 | 0.704 ±0.007 | 1.384 ±0.167 |
| | | analytic | 0.17 | 0.345 ±0.051 | 0.930 ±0.093 | 0.498 ±0.038 | 0.426 ±0.133 | 0.658 ±0.048 |
| | | validation-selected | 0.21 | 0.352 ±0.034 | 0.918 ±0.085 | 0.506 ±0.026 | 0.459 ±0.085 | 0.639 ±0.062 |
| Breast Cancer Wisconsin | Jev, zero-shot | default | 0.50 | 0.382 ±0.007 | 0.993 ±0.012 | 0.552 ±0.008 | 0.405 ±0.014 | 0.605 ±0.030 |
| | | analytic | 0.17 | 0.369 ±0.001 | 1.000 ±0.000 | 0.539 ±0.001 | 0.370 ±0.004 | 0.630 ±0.004 |
| | | validation-selected | 0.77 | 0.605 ±0.079 | 0.912 ±0.083 | 0.721 ±0.040 | 0.735 ±0.068 | 0.395 ±0.089 |
| | Qwen3.6-27B, zero-shot | default | 0.50 | 0.991 ±0.019 | 0.531 ±0.056 | 0.690 ±0.048 | 0.825 ±0.021 | 0.866 ±0.104 |
| | | analytic | 0.17 | 0.838 ±0.031 | 0.905 ±0.032 | 0.870 ±0.023 | 0.900 ±0.018 | 0.240 ±0.059 |
| | | validation-selected | 0.15 | 0.789 ±0.058 | 0.926 ±0.046 | 0.850 ±0.027 | 0.878 ±0.029 | 0.231 ±0.059 |
| | XGBoost, supervised | default | 0.50 | 0.942 ±0.023 | 0.919 ±0.047 | 0.930 ±0.030 | 0.949 ±0.021 | 0.170 ±0.088 |
| | | analytic | 0.17 | 0.879 ±0.036 | 0.960 ±0.046 | 0.917 ±0.029 | 0.936 ±0.022 | 0.124 ±0.084 |
| | | validation-selected | 0.27 | 0.895 ±0.042 | 0.943 ±0.053 | 0.917 ±0.026 | 0.937 ±0.019 | 0.147 ±0.089 |
| | BERT, supervised | default | 0.50 | 0.613 ±0.333 | 0.593 ±0.334 | 0.595 ±0.321 | 0.778 ±0.092 | 0.822 ±0.574 |
| | | analytic | 0.17 | 0.560 ±0.127 | 0.879 ±0.080 | 0.671 ±0.089 | 0.661 ±0.161 | 0.518 ±0.127 |
| | | validation-selected | 0.20 | 0.587 ±0.103 | 0.845 ±0.079 | 0.687 ±0.079 | 0.709 ±0.099 | 0.519 ±0.168 |
| SST-2 | Jev, zero-shot | default | 0.50 | 0.979 | 0.941 | 0.960 | 0.960 | 0.040 |
| | | analytic | 0.50 | 0.979 | 0.941 | 0.960 | 0.960 | 0.040 |
| | | validation-selected | 0.36 | 0.975 | 0.959 | 0.967 | 0.967 | 0.033 |
| | Qwen3.6-27B, zero-shot | default | 0.50 | 0.981 | 0.932 | 0.956 | 0.956 | 0.044 |
| | | analytic | 0.50 | 0.981 | 0.932 | 0.956 | 0.956 | 0.044 |
| | | validation-selected | 0.29 | 0.972 | 0.953 | 0.962 | 0.962 | 0.038 |
| | XGBoost, supervised | default | 0.50 | 0.781 | 0.820 | 0.800 | 0.791 | 0.209 |
| | | analytic | 0.50 | 0.781 | 0.820 | 0.800 | 0.791 | 0.209 |
| | | validation-selected | 0.55 | 0.798 | 0.800 | 0.799 | 0.795 | 0.205 |
| | BERT, supervised | default | 0.50 | 0.901 | 0.939 | 0.920 | 0.916 | 0.084 |
| | | analytic | 0.50 | 0.901 | 0.939 | 0.920 | 0.916 | 0.084 |
| | | validation-selected | 0.49 | 0.901 | 0.939 | 0.920 | 0.916 | 0.084 |

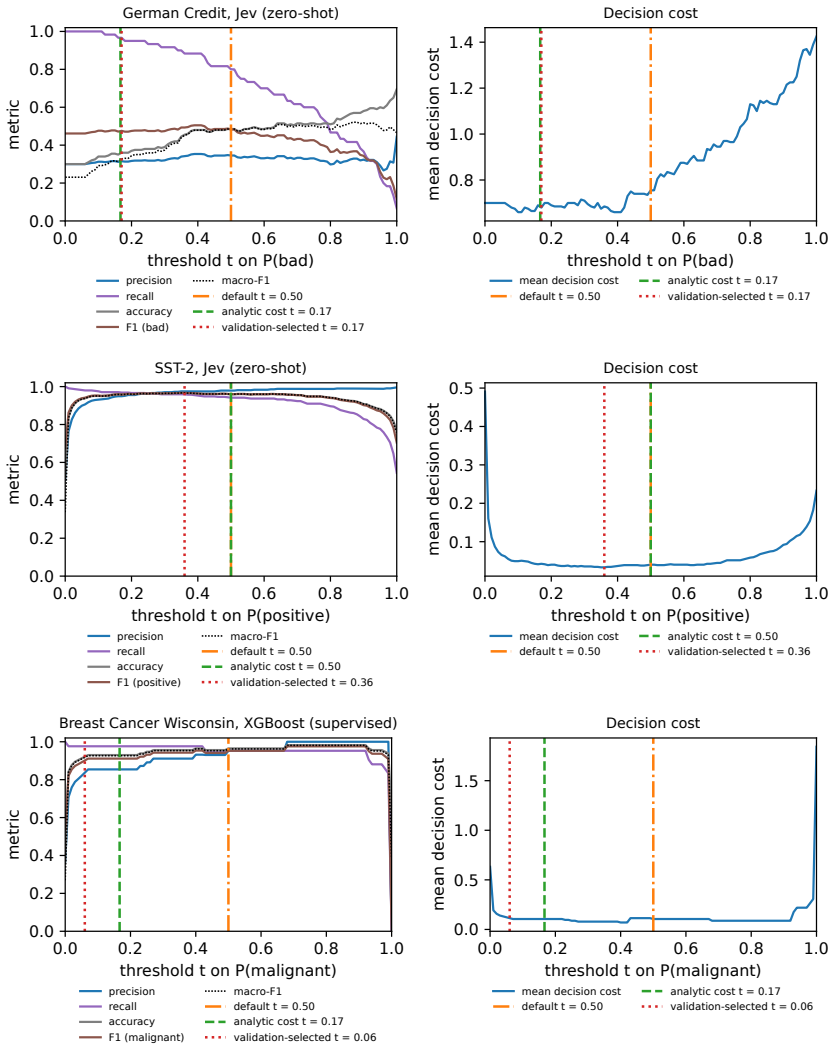



Figure 6: Threshold sweeps (first split repeat) for Jev on German Credit and SST-2, and for XGBoost on Breast Cancer Wisconsin Diagnostic. For SST-2 the costs are symmetric, so the analytic threshold coincides with the default.

# G Importance sampling in full

For each dataset the pool is the fixed set of examples with cached Jev predictions. The sampling procedure sees only those predictions: it draws $n$ items with replacement and asks for the labels of the drawn items only. The Hansen–Hurwitz estimate of the error rate is $\frac{1}{n}\sum_j \ell_{i_j}/(Nq_{i_j})$, which is unbiased for any positive $q$, and accuracy is one minus it. We run 1,000 seeded trials at each budget $n \in \{25, 50, 100, 200, 400\}$ and compare with the full-pool accuracy, which is used only as ground truth.

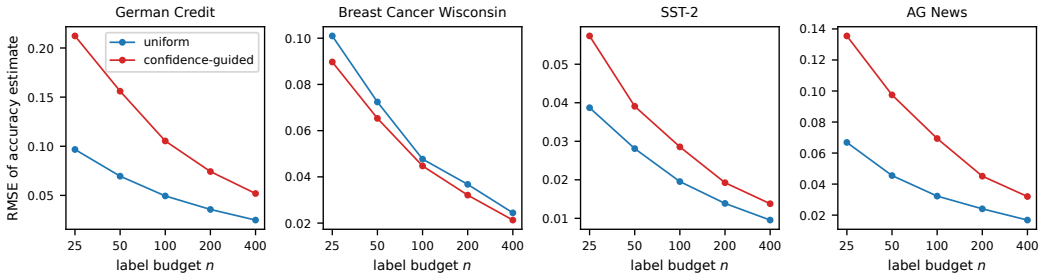



Figure 7: RMSE of the accuracy estimate against the label budget.

Two things explain the result. First, because of the rounding, a large part of each text pool has a reported uncertainty of exactly zero: 539 of 872 items on SST-2 and 5,621 of 7,600 on AG News. Those items receive only the minimum sampling probability $\varepsilon/N$, yet they contain 5 of the 35 errors on SST-2 and 307 of 923 on AG News. Second, for a 0/1 loss a proposal proportional to the expected loss has the same variance as uniform sampling under perfect calibration, and the optimum is proportional to the square root of the expected loss. With $\alpha = 0.5$ the ratios at $n = 400$ are 1.17 on Breast Cancer Wisconsin Diagnostic, 0.32 on German Credit, 0.52 on SST-2 and 0.25 on AG News: still below 1 on 3 of the 4. Here the check says to sample uniformly unless the reported uncertainties are first floored or recalibrated.

## H Explanations in full

KernelSHAP uses 200 coalitions with the all-removed input as background, and LIME 200 samples. Text examples were drawn among those with 6 to 40 words (835 of 872 for SST-2, 4,979 of 7,600 for AG News), so that deleting five words is meaningful. Faithfulness is tested by deleting the top-$k$ attributed units and re-querying Jev, against deleting $k$ random units (5 draws). KernelSHAP's attributions plus its base value reconstruct Jev's probability exactly (largest additivity error 4e-16). The two methods agree moderately to well on German Credit and SST-2 ($\tau = 0.71$ and 0.51) and little on Breast Cancer Wisconsin Diagnostic and AG News ($\tau = 0.37$ and 0.19), where LIME's linear surrogate fits poorly ($R^2 = 0.54$ and 0.29, against 0.80 on German Credit). With every feature removed, Jev assigns the positive class a probability of 0.69 on German Credit and 0.51 on Breast Cancer Wisconsin Diagnostic, where the true shares are 0.30 and 0.37. Perturbation methods are paid for in model calls: at most 423 per example here, and 29,718 requests in total (USD 0.54), between 6,666 and 7,969 per dataset after caching repeated inputs.

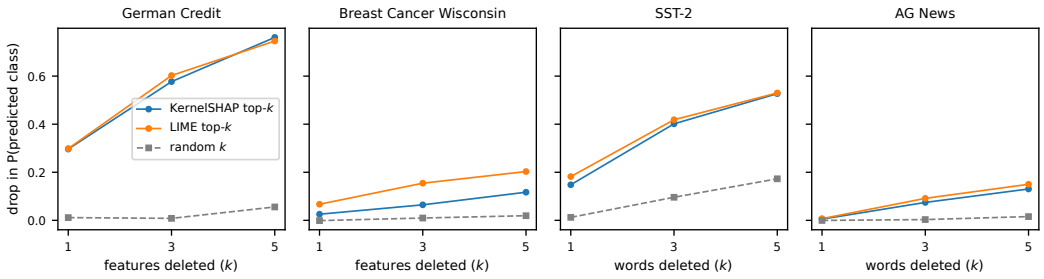



Figure 8: Deletion test: mean drop in Jev's predicted-class probability after deleting the $k$ top-attributed units (KernelSHAP, LIME) or $k$ random units.

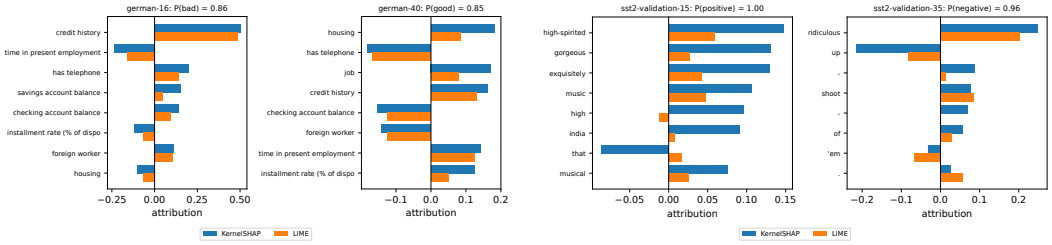



Figure 9: KernelSHAP and LIME attributions for two German Credit applicants (left) and two SST-2 sentences (right): the eight units with the largest absolute KernelSHAP value.

# I Exact requests and responses

One complete exchange per dataset, copied from the request log. The example is the first evaluation example, in a seeded order, whose request is in the log. The request and response bodies are the JSON that crossed the wire, re-indented for reading; non-ASCII characters are shown as JSON escapes. Headers are not recorded, so the API key cannot appear.

The 100 requests made before the log existed are not shown, because there is nothing to copy. For 99 of them the response was cached in parsed form; regenerating their requests from code and hashing them reproduces 99 of the 99 stored cache keys, which shows that the same input, instructions, label descriptions and model version were sent, but does not recover the payloads or the raw responses. The remaining one was a connectivity check that was not cached.

## German Credit

Example `german-870`, gold label `good`; ledger number 951, sent 2026-09-30T13:43:22.039+00:00; `POST https://api.typesafe.ai/v1/systemone`; SDK `typesafe-sdk 0.7.2`; requested model `jev-1.13.0`, reported model `jev-1.13.0`; HTTP status 200; latency 0.18 s.

### Request body as sent

```
{
 "state": "checking account balance: no checking account\nloan duration (months): 36\ncredit history:
     ↪ delayed previously\nloan purpose: furniture/equipment\ncredit amount (DM): 7678\nsavings account
     ↪ balance: 500 to 1000 DM\ntime in present employment: 4 to 7 years\ninstallment rate (% of
     ↪ disposable income): 2\npersonal status and sex: female div/dep/mar\nother debtors / guarantors:
     ↪ none\nyears at present residence: 4\nmost valuable property: car\nage (years): 40\nother
     ↪ installment plans: none\nhousing: own\nnumber of existing credits at this bank: 2\njob: skilled\
     ↪ nnumber of dependents: 1\nhas telephone: yes\nforeign worker: yes",
 "model": "jev-1.13.0",
 "questions": {
  "label": {
   "type": "choice",
   "instructions": "Assess the credit risk of a bank loan applicant described below. Fields that are not
     ↪ listed are unknown.",
   "criteria": {
    "good": "The applicant is a good credit risk and is likely to repay the loan.",
    "bad": "The applicant is a bad credit risk and is likely to default on the loan."
   }
  }
 }
}
```

### Response body as received

```
{
 "model": "jev-1.13.0",
 "answers": {
  "label": {
   "type": "choice",
   "choice": "good",
   "confidence": 0.13,
   "probabilities": {
    "good": 0.56,
    "bad": 0.44
   }
```

```
  }
 },
 "usage": {
  "input_tokens": 520,
  "output_tokens": 31
 }
}
```

### Breast Cancer Wisconsin Diagnostic

Example `wdbc-360`, gold label `benign`; ledger number 1423, sent 2026-09-30T13:43:34.432+00:00; `POST https://api.typesafe.ai/v1/systemone`; SDK `typesafe-sdk 0.7.2`; requested model `jev-1.13.0`, reported model `jev-1.13.0`; HTTP status 200; latency 0.11 s.

#### Request body as sent

```
{
 "state": "mean radius: 12.54\nmean texture: 18.07\nmean perimeter: 79.42\nmean area: 491.9\nmean
     ↪ smoothness: 0.07436\nmean compactness: 0.0265\nmean concavity: 0.001194\nmean concave points:
     ↪ 0.005449\nmean symmetry: 0.1528\nmean fractal dimension: 0.05185\nradius error: 0.3511\ntexture
     ↪ error: 0.9527\nperimeter error: 2.329\narea error: 28.3\nsmoothness error: 0.005783\ncompactness
     ↪ error: 0.004693\nconcavity error: 0.0007929\nconcave points error: 0.003617\nsymmetry error:
     ↪ 0.02043\nfractal dimension error: 0.001058\nworst radius: 13.72\nworst texture: 20.98\nworst
     ↪ perimeter: 86.82\nworst area: 585.7\nworst smoothness: 0.09293\nworst compactness: 0.04327\nworst
     ↪  concavity: 0.003581\nworst concave points: 0.01635\nworst symmetry: 0.2233\nworst fractal
     ↪ dimension: 0.05521",
 "model": "jev-1.13.0",
 "questions": {
  "label": {
   "type": "choice",
   "instructions": "Assess whether the breast mass described by the cell-nucleus measurements below is
     ↪ benign or malignant. Fields that are not listed are unknown.",
   "criteria": {
    "benign": "The mass is benign (not cancerous).",
    "malignant": "The mass is malignant (cancerous)."
   }
  }
 }
}
```

#### Response body as received

```
{
 "model": "jev-1.13.0",
 "answers": {
  "label": {
   "type": "choice",
   "choice": "benign",
   "confidence": 0.06,
   "probabilities": {
    "malignant": 0.47,
    "benign": 0.53
   }
  }
 },
 "usage": {
  "input_tokens": 733,
  "output_tokens": 35
 }
}
```

### SST-2

Example `sst2-validation-36`, gold label `positive`; ledger number 1668, sent 2026-09-30T13:43:49.101+00:00; `POST https://api.typesafe.ai/v1/systemone`; SDK `typesafe-sdk 0.7.2`; requested model `jev-1.13.0`, reported model `jev-1.13.0`; HTTP status 200; latency 0.13 s.

#### Request body as sent

```
{
 "state": "the weight of the piece , the unerring professionalism of the chilly production , and the
     ↪ fascination embedded in the lurid topic prove recommendation enough .",
```

```
 "model": "jev-1.13.0",
 "questions": {
  "label": {
   "type": "choice",
   "instructions": "Classify the sentiment of the movie review sentence below.",
   "criteria": {
    "negative": "The sentence expresses a negative opinion of the movie.",
    "positive": "The sentence expresses a positive opinion of the movie."
   }
  }
 }
}
```

### Response body as received

```
{
 "model": "jev-1.13.0",
 "answers": {
  "label": {
   "type": "choice",
   "choice": "positive",
   "confidence": 0.96,
   "probabilities": {
    "negative": 0.02,
    "positive": 0.98
   }
  }
 },
 "usage": {
  "input_tokens": 359,
  "output_tokens": 31
 }
}
```

## AG News

Example `agnews-test-3752`, gold label `Business`; ledger number 9835, sent 2026-09-30T13:46:45.754+00:00; `POST https://api.typesafe.ai/v1/systemone`; SDK `typesafe-sdk 0.7.2`; requested model `jev-1.13.0`, reported model `jev-1.13.0`; HTTP status 200; latency 0.11 s.

### Request body as sent

```
{
 "state": "McTeer: Lonesome Dove to be an Aggie NEW YORK (CNN/Money) - A New Economy champion, a lover of
     ↪  the Texas picker poets who write lovesick country songs...and, oh, by the way, a member of the
     ↪ Federal Reserve system for 36 years.",
 "model": "jev-1.13.0",
 "questions": {
  "label": {
   "type": "choice",
   "instructions": "Classify the topic of the news article below.",
   "criteria": {
    "World": "International news, politics, conflicts and world affairs.",
    "Sports": "Sports news: games, athletes, teams and competitions.",
    "Business": "Business and economic news: companies, markets, finance and the economy.",
    "Sci/Tech": "Science and technology news: computing, the internet, research, space and health science
     ↪ ."
   }
  }
 }
}
```

### Response body as received

```
{
 "model": "jev-1.13.0",
 "answers": {
  "label": {
   "type": "choice",
   "choice": "Business",
   "confidence": 0.97,
   "probabilities": {
    "Sci/Tech": 0.0,
    "Business": 0.98,
    "Sports": 0.01,
```

```
    "World": 0.01
   }
  }
 },
 "usage": {
  "input_tokens": 446,
  "output_tokens": 47
 }
}
```


# J Qwen prompts

One prompt per dataset as rendered by Qwen's own tokenizer inside the job, with thinking disabled, and the distribution obtained by scoring every option after it. The end-of-turn token is part of every scored option, which makes an option that is a prefix of another a complete answer.

### German Credit

Example `german-529`. Prompt with thinking disabled, as rendered by the model's tokenizer inside the job:

```
<|im_start|>system
Assess the credit risk of a bank loan applicant described below. Fields that are not listed are unknown.

Options:
- good: The applicant is a good credit risk and is likely to repay the loan.
- bad: The applicant is a bad credit risk and is likely to default on the loan.

Answer with exactly one option name.<|im_end|>
<|im_start|>user
checking account balance: below 0 DM
loan duration (months): 6
credit history: existing paid
loan purpose: new car
credit amount (DM): 662
savings account balance: below 100 DM
time in present employment: less than 1 year
installment rate (% of disposable income): 3
personal status and sex: male single
other debtors / guarantors: none
years at present residence: 4
most valuable property: real estate
age (years): 41
other installment plans: none
housing: own
number of existing credits at this bank: 1
job: unskilled resident
number of dependents: 2
has telephone: yes
foreign worker: yes<|im_end|>
<|im_start|>assistant
<think>

</think>
```

Qwen3.6-27B's distribution: `good` 0.679, `bad` 0.321.

### Breast Cancer Wisconsin Diagnostic

Example `wdbc-477`. Prompt with thinking disabled, as rendered by the model's tokenizer inside the job:

```
<|im_start|>system
Assess whether the breast mass described by the cell-nucleus measurements below is benign or malignant.
    ↪ Fields that are not listed are unknown.

Options:
- benign: The mass is benign (not cancerous).
- malignant: The mass is malignant (cancerous).

Answer with exactly one option name.<|im_end|>
<|im_start|>user
```

```
mean radius: 13.9
mean texture: 16.62
mean perimeter: 88.97
mean area: 599.4
mean smoothness: 0.06828
mean compactness: 0.05319
mean concavity: 0.02224
mean concave points: 0.01339
mean symmetry: 0.1813
mean fractal dimension: 0.05536
radius error: 0.1555
texture error: 0.5762
perimeter error: 1.392
area error: 14.03
smoothness error: 0.003308
compactness error: 0.01315
concavity error: 0.009904
concave points error: 0.004832
symmetry error: 0.01316
fractal dimension error: 0.002095
worst radius: 15.14
worst texture: 21.8
worst perimeter: 101.2
worst area: 718.9
worst smoothness: 0.09384
worst compactness: 0.2006
worst concavity: 0.1384
worst concave points: 0.06222
worst symmetry: 0.2679
worst fractal dimension: 0.07698<|im_end|>
<|im_start|>assistant
<think>

</think>
```

Qwen3.6-27B's distribution: `benign` 0.977, `malignant` 0.023.

### SST-2

Example `sst2-validation-2`. Prompt with thinking disabled, as rendered by the model's tokenizer inside the job:

```
<|im_start|>system
Classify the sentiment of the movie review sentence below.

Options:
- negative: The sentence expresses a negative opinion of the movie.
- positive: The sentence expresses a positive opinion of the movie.

Answer with exactly one option name.<|im_end|>
<|im_start|>user
allows us to hope that nolan is poised to embark a major career as a commercial yet inventive filmmaker
    ↪ .<|im_end|>
<|im_start|>assistant
<think>

</think>
```

Qwen3.6-27B's distribution: `positive` 1.000, `negative` 0.000.

### AG News

Example `agnews-test-20`. Prompt with thinking disabled, as rendered by the model's tokenizer inside the job:

```
<|im_start|>system
Classify the topic of the news article below.

Options:
- World: International news, politics, conflicts and world affairs.
- Sports: Sports news: games, athletes, teams and competitions.
- Business: Business and economic news: companies, markets, finance and the economy.
- Sci/Tech: Science and technology news: computing, the internet, research, space and health science.

Answer with exactly one option name.<|im_end|>
```

```
<|im_start|>user
IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest
    ↪ headcount since 1991.<|im_end|>
<|im_start|>assistant
<think>

</think>
```

Qwen3.6-27B's distribution: `Business` 0.997, `Sci/Tech` 0.001, `World` 0.001, `Sports` 0.001.

# K Dataset documentation

Evaluation identifiers were frozen on 2026-09-30T13:20:53Z (UTC), before any template check, training or threshold selection; every stage verifies their hash before running.

Table 17: Datasets and split protocol. "Jev requests" is the number of examples Jev predicts: every row of the tabular datasets (each repeat's validation and test parts both use its predictions), and the held-out split of the text datasets plus, for SST-2, 500 validation items for threshold selection.

| Dataset | Type | Classes | Evaluation examples | Train / validation | Jev requests |
|---|---|---|---|---|---|
| German Credit | tabular | 2 | 10 stratified repeats, test 200 each | 600 / 200 per repeat | 1,000 |
| Breast Cancer Wisconsin Diagnostic | tabular | 2 | 10 stratified repeats, test 114 each | 341 / 114 per repeat | 569 |
| SST-2 | text | 2 | official validation split, 872 | 65,349 / 2,000 | 1,372 |
| AG News | text | 4 | official test split, 7,600 | 115,000 / 5,000 | 7,600 |

Table 17 gives the split protocol in full.

**German Credit.** *Source:* OpenML dataset 31 (credit-g, version 1). *Classes, in the order sent:* `good` (The applicant is a good credit risk and is likely to repay the loan., 621 evaluation examples); `bad` (The applicant is a bad credit risk and is likely to default on the loan., 264 evaluation examples). *Positive class:* bad. *Splits:* 10 stratified repeats with seeds 0–9; train 600, validation 200, test 200 in each; 885 distinct rows are a test row in some repeat. *Input:* one `name: value` line per feature (20 features), categorical codes decoded; a missing value is written as `unknown` (0 missing values in this dataset). *Instructions:* "Assess the credit risk of a bank loan applicant described below. Fields that are not listed are unknown."

**Breast Cancer Wisconsin Diagnostic.** *Source:* scikit-learn 1.9.1 sklearn.datasets.load_breast_cancer (UCI Machine Learning Repository, Breast Cancer Wisconsin (Diagnostic), 569 rows, 30 features). *Classes, in the order sent:* `benign` (The mass is benign (not cancerous)., 317 evaluation examples); `malignant` (The mass is malignant (cancerous)., 189 evaluation examples). *Positive class:* malignant. *Splits:* 10 stratified repeats with seeds 0–9; train 341, validation 114, test 114 in each; 506 distinct rows are a test row in some repeat. *Input:* one `name: value` line per feature (30 features), categorical codes decoded; a missing value is written as `unknown` (0 missing values in this dataset). *Instructions:* "Assess whether the breast mass described by the cell-nucleus measurements below is benign or malignant. Fields that are not listed are unknown."

**SST-2.** *Source:* Hugging Face nyu-mll/glue, configuration sst2, revision bcdcba79d07bc864c1c254ccfcedcce55bcc9a8c. *Classes, in the order sent:* `negative` (The sentence expresses a negative opinion of the movie., 428 evaluation examples); `positive` (The sentence expresses a positive opinion of the movie., 444 evaluation examples). *Positive class:* positive. *Splits:* official validation split (872 examples, all used); validation 2,000 drawn stratified from the official training split with seed 0; training 65,349. *Input:* the text itself, with whitespace runs collapsed to one space; leading and trailing space removed; no example is empty. *Instructions:* "Classify the sentiment of the movie review sentence below."

**AG News.** *Source:* Hugging Face fancyzhx/ag_news, revision eb185aade064a813bc0b7f42de02595523103ca4. *Classes, in the order sent:* `World` (International news, politics, conflicts and world affairs., 1900 evaluation examples); `Sports` (Sports news: games, athletes, teams and competitions., 1900 evaluation examples); `Business` (Business and economic news: companies, markets,

finance and the economy., 1900 evaluation examples); `Sci/Tech` (Science and technology news: computing, the internet, research, space and health science., 1900 evaluation examples). *Positive class:* not applicable. *Splits:* official test split (7,600 examples, all used); validation 5,000 drawn stratified from the official training split with seed 0; training 115,000. *Input:* the text itself, with whitespace runs collapsed to one space; leading and trailing space removed; no example is empty. *Instructions:* "Classify the topic of the news article below."

# L Per-class metrics

| | | Jev | | | Qwen3.6-27B | | | XGBoost | | | BERT | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Class | Support | P | R | F1 | P | R | F1 | P | R | F1 | P | R | F1 |
| good | 1400 | 0.78 | 0.33 | 0.46 | 0.67 | 0.67 | 0.67 | 0.78 | 0.88 | 0.83 | 0.71 | 0.97 | 0.82 |
| bad | 600 | 0.33 | 0.79 | 0.47 | 0.24 | 0.24 | 0.24 | 0.60 | 0.41 | 0.49 | 0.55 | 0.09 | 0.16 |

Table 18: Per-class precision (P), recall (R) and F1 on German Credit. Support and metrics are pooled over the split repeats.

| | | Jev | | | Qwen3.6-27B | | | XGBoost | | | BERT | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Class | Support | P | R | F1 | P | R | F1 | P | R | F1 | P | R | F1 |
| benign | 720 | 0.94 | 0.07 | 0.13 | 0.78 | 1.00 | 0.88 | 0.95 | 0.97 | 0.96 | 0.79 | 0.89 | 0.83 |
| malignant | 420 | 0.38 | 0.99 | 0.55 | 0.99 | 0.53 | 0.69 | 0.94 | 0.92 | 0.93 | 0.75 | 0.59 | 0.66 |

Table 19: Per-class precision (P), recall (R) and F1 on Breast Cancer Wisconsin Diagnostic. Support and metrics are pooled over the split repeats.

| | | Jev | | | Qwen3.6-27B | | | XGBoost | | | BERT | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Class | Support | P | R | F1 | P | R | F1 | P | R | F1 | P | R | F1 |
| negative | 428 | 0.94 | 0.98 | 0.96 | 0.93 | 0.98 | 0.96 | 0.80 | 0.76 | 0.78 | 0.93 | 0.89 | 0.91 |
| positive | 444 | 0.98 | 0.94 | 0.96 | 0.98 | 0.93 | 0.96 | 0.78 | 0.82 | 0.80 | 0.90 | 0.94 | 0.92 |

Table 20: Per-class precision (P), recall (R) and F1 on SST-2.

| | | Jev | | | Qwen3.6-27B | | | XGBoost | | | BERT | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Class | Support | P | R | F1 | P | R | F1 | P | R | F1 | P | R | F1 |
| World | 1900 | 0.90 | 0.89 | 0.90 | 0.92 | 0.87 | 0.90 | 0.93 | 0.91 | 0.92 | 0.97 | 0.93 | 0.95 |
| Sports | 1900 | 0.97 | 0.98 | 0.97 | 0.96 | 0.99 | 0.97 | 0.95 | 0.98 | 0.96 | 0.99 | 0.99 | 0.99 |
| Business | 1900 | 0.76 | 0.92 | 0.83 | 0.75 | 0.92 | 0.83 | 0.89 | 0.88 | 0.88 | 0.91 | 0.92 | 0.91 |
| Sci/Tech | 1900 | 0.92 | 0.72 | 0.81 | 0.90 | 0.74 | 0.81 | 0.89 | 0.89 | 0.89 | 0.90 | 0.93 | 0.92 |

Table 21: Per-class precision (P), recall (R) and F1 on AG News.